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Indian Statistical Institute
B. Statistical Data Science II (2025-26)
IVth Semester Examination 2026

Paper: Mathematics IV
Time : $1\frac{1}{2}$ Hrs.

Part I

Date: 18/05/2026
Full Marks : 25

Note: Symbols used have their usual meaning. *The figures in the margin indicate full marks.*

Answer any FIVE from the following questions.

5 × 5 = 25

1. (a) Prove or disprove that a vector function $\mathbf{a}(t)$ has a constant direction if and only if

$$\mathbf{a} \times \frac{d\mathbf{a}}{dt} = \mathbf{0}.$$

- (b) Find the equation of the plane passing through the point with position vector $\mathbf{a}$, perpendicular to the plane $\mathbf{r} \cdot \mathbf{n} = q$, and parallel to the line $\mathbf{r} = \mathbf{b} + t\mathbf{c}$.

[2 + 3]

2. If

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})},$$

then prove that

$$\mathbf{a} = \frac{\mathbf{b}' \times \mathbf{c}'}{\mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}')}, \quad \mathbf{b} = \frac{\mathbf{c}' \times \mathbf{a}'}{\mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}')}, \quad \mathbf{c} = \frac{\mathbf{a}' \times \mathbf{b}'}{\mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}')}.$$

[5]

3. (a) For twice differentiable scalar-valued functions f and g , prove that

$$\nabla \cdot (f \nabla g) - \nabla \cdot (g \nabla f) = f \nabla^2 g - g \nabla^2 f.$$

- (b) Determine whether the vector field

$$\mathbf{F} = (e^x \cos y + e^{-x} \sin z) \hat{i} - e^x \sin y \hat{j} + e^{-x} \cos z \hat{k}$$

can be expressed as the gradient of a C^2 -function (i.e., function with continuous second derivatives).

[3 + 2]

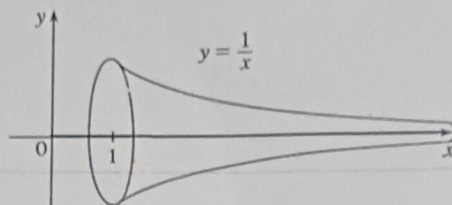
4. (a) Find the scale factors and the corresponding unit vectors for the paraboloidal coordinate system (u, v, ϕ) defined by

$$x = uv \cos \phi, \quad y = uv \sin \phi \quad \text{and} \quad z = \frac{1}{2}(u^2 - v^2).$$

- (b) Derive the expressions for the gradient of a scalar function f and the divergence of a vector field $\mathbf{a}$ in the above paraboloidal coordinate system.

[2 + (1 + 2)]

5. (a) Find a parametric representation of the intersection of $2x - y + 3z = 2$ and $x + 2y - z = 3$.
 (b) If the region $\mathcal{R} = \{(x, y) \mid x \geq 1, 0 \leq y \leq 1/x\}$ is rotated about the x -axis, then show that the volume of the resulting space as shown in the figure is finite where as the surface area is infinite.



[2 + 3]

6. (a) Prove that the arc length over any interval for the function

$$f(x) = \frac{1}{4}(e^x - e^{-x}),$$

is equal to the area under the curve over the same interval.

- (b) Find the radius of curvature for $\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$ at the points where it touches the coordinate axes.

[2½ + 2½]

7. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F} = (x + y)\mathbf{i} + (2x - z)\mathbf{j} + (y + z)\mathbf{k}$$

and C is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$ and orientation $(2, 0, 0) \rightarrow (0, 3, 0) \rightarrow (0, 0, 6)$.

[5]

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Answer any FIVE from the following questions.

5 × 5 = 25

1. Verify Stokes' theorem for the vector field $\mathbf{F} = (y^2 + z^2 - x^2)\mathbf{i} + (z^2 + x^2 - y^2)\mathbf{j} + (x^2 + y^2 - z^2)\mathbf{k}$ over the portion of the surface

$$x^2 + y^2 + z^2 - 2ax - 2cz = 0, \quad z \geq 0.$$

[5]

2. (a) Find the largest interval $|x| < h$ on which $\frac{dy}{dx} = 1 + y^2$, $y(0) = 0$, has unique solution.
(b) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function defined by $f(x, y) = e^{x+y} + y + \lambda x$. Determine the value(s) of λ for which there exists a unique implicit function $y = \phi(x)$ satisfying $f(x, y) = 0$ in a neighborhood of $\left(\frac{1}{2}, -\frac{1}{2}\right)$.

[2 + 3]

3. Examine whether f, g and h are functionally related

$$f(x, y, z, w) = x + y + z + w,$$

$$g(x, y, z, w) = x^2 + y^2 + w^2,$$

$$h(x, y, z, w) = 2(xy + xz + xw + yz + yw + zw).$$

[5]

4. Find the general solution of the nonhomogeneous linear system

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -10 & 6 \\ -12 & 7 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 10e^{-3t} \\ 18e^{-3t} \end{pmatrix}.$$

[5]

5. Find the solution of the following system of equations

$$\frac{dx}{dt} - 4x + 2y = 2t,$$

$$\frac{dy}{dt} - 8x + 4y = 1,$$

$$x(0) = 3, \quad y(0) = 5,$$

using Laplace transformation.

[5]

6. Given a solution $\varphi_1(x) = \frac{\cos x}{x^2}$ of

$$x^2 y'' + 4xy' + (2 + x^2)y = 0,$$

find the general solution of

$$x^2 y'' + 4xy' + (2 + x^2)y = x^2 \quad (x > 0).$$

[5]

7. Show that the differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

has an integrating factor $\mu(x)$, which is a function of x alone if and only if

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$$

is a function of x only, say $\phi(x)$, and in this case $\mu(x)$ must be a constant multiple of $e^{\int \phi(x) dx}$.

[5]